

Class 6: Intro to Proofs

Goals:

1. Know what type of statements you can prove by giving an example or disprove by giving a counterexample.
2. Be able to construct a direct proof for an implication.

One way to prove an existential statement is true is by providing an example

The Mean Value Theorem in calculus is an example of an existence theorem that cannot be proven by an example.

An example of a universal statement is “Every integer greater than 100 is even.”

Is this universal statement true?

How can you disprove a universal statement?

Starting facts and definitions

Natural numbers:

Whole numbers:

Integers:

Rational numbers:

Real numbers:

Irrational numbers:

Even and odd numbers:

Two basic facts we will use in homework.

Parity Theorem: Every integer is either even or odd but not both.

Zero Product Property: If $x \cdot y = 0$ then either $x = 0$ or $y = 0$.

Direct Proofs: Most theorems have the logical structure of an implication, $p \rightarrow q$. If certain conditions (premises) are met, the conclusion follows. A direct proof of the theorem starts by assuming the antecedent, p , is true and shows that the consequent, q , follows.

Example: Give a direct proof that the cube of an even integer is even.

What you should write:

Prove that the cube of an even integer is even.

Proof: Let x be any even integer. [Show x^3 is even.]

Since x is even, $x = 2k$ for some integer k .

This means $x^3 = (2k)^3 = 8k^3 = 2(4k^3)$ is even since $4k^3$ is an integer whenever k is an integer.

More practice

1. Prove that the sum of two odd integers is even.
2. Prove that the product of two rational numbers is rational.